L7-1 Resampling statistics

General procedure for confidence intervals

- Use an **estimate of the parameter** $\hat{\theta}$ and the **standard error** (SE).
- To get a 95% confidence interval:
  $[\hat{\theta} \pm 1.96\, SE]$
- Or, more general an α confidence interval:
  $[\hat{\theta} \pm z_{1-\frac{\alpha}{2}}\, SE]$
- What to do when the SE is not known?
  Traditional solution: the *Delta method*, which builds on a Taylor expansion of the expression to approximate the SE.

```
library(msm)
deltamethod(…)
```

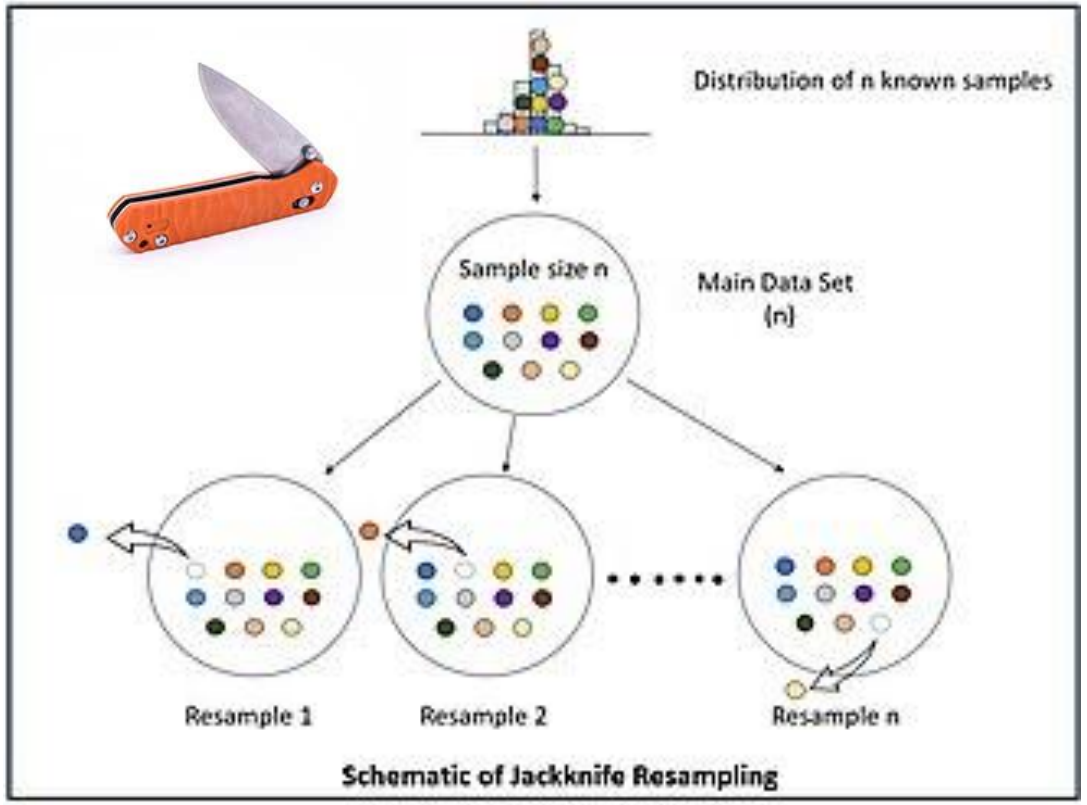
Jackson 2024

Resampling statistics

- Alternative methods for obtaining sampling distributions of statistics. For example, confidence intervals of estimates, standard errors.
- Three types:
  - Jack-knife
  - Bootstrap
  - Permutation tests

Jack-knife

- An early approach to resampling.
- Similar to leave-1-out validation in machine learning.
- Deterministic process (produces the same result every time).
- Gives an estimate of the standard error (SE) and confidence interval of, *e.g.*, an estimate of the mean.
- Several packages in R:
  `library(boot), library(bootstrap)`



R

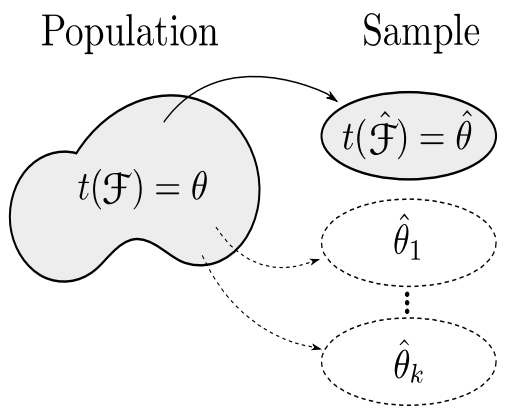
```
library(bootstrap)
dat <- rnorm(n, mean, sd)
theta <- function(dat){mean(dat)}
x_jk <- jackknife(dat, theta) # e.g., CIs
```

Efron & Tibshirani 1986

Bootstrapping

- More computationally demanding.
- More flexible, can be adapted to most estimation problems.
- Non-deterministic process (slightly different results every time).
- Dependent on the number of samples.
- Basic principle: random resampling with replacement from the original sample.
- Estimators derived from the subsamples.





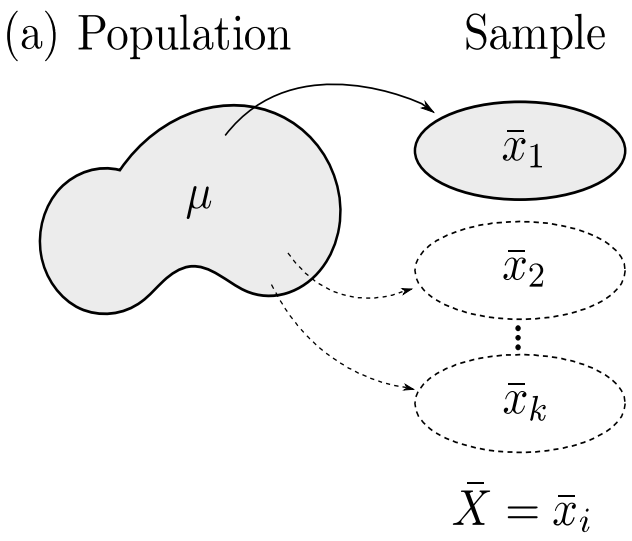
R

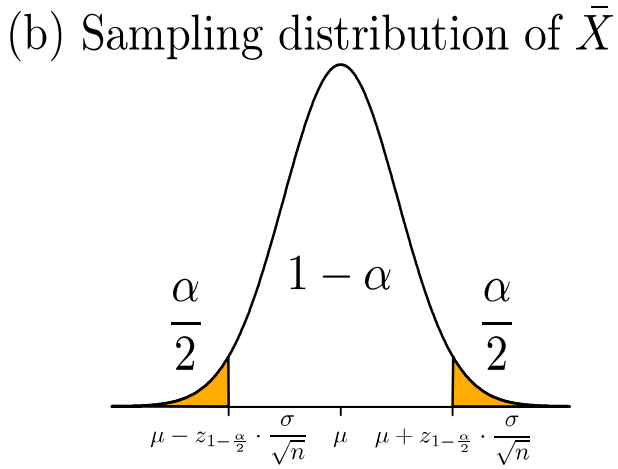
```
library(bootstrap)
…
boot1 <- bootstrap(dat, nboot, theta) # e.g., Cis
ci95 <- quantile(boot1$thetastar,c(0.025,0.975))
```

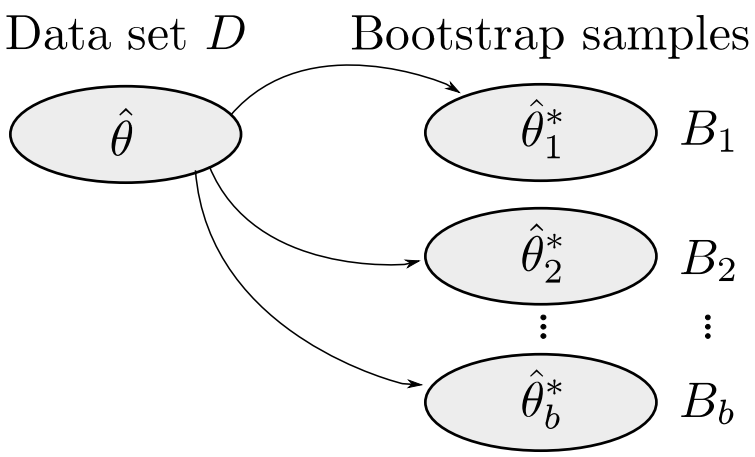
Efron & Tibshirani 1993

Bootstrapping

- Simple example: estimating the mean by resampling.
- Main point: can be used for various other statistics θ.







Non-parametric bootstrapping

1. Start with a dataset *D*.
2. Compute an estimate $\hat{\theta}$ of the parameter θ from *D*.
3. Make a bootstrap set *B* of size *n* by resampling *D* with replacement.
4. Repeat to make *b* bootstrap sets B_1, B_2,… B_b
5. Calculate $\widehat{\theta_i}$ for each bootstrap set B_i
6. This gives a *bootstrap distribution* of all $\widehat{\theta_i}$

- To obtain a 95% **confidence interval**:
- Use the 2.5 and 97.5 percentiles from the bootstrap distribution.

```
packages: bootstrap, boot and bcaboot
quantile (Y, c(0.025, 0.975))
```

Permutation tests (exact test)

- Non-parametric tests applicable to any distribution.
- Basic idea: Go through *all possible rearrangements* of the observed data.
- May require very heavy computations, in particular with large datasets.
- Simple special case: Fisher's exact test.

See Lecture L1-6 Nonparametric Statistical Tests